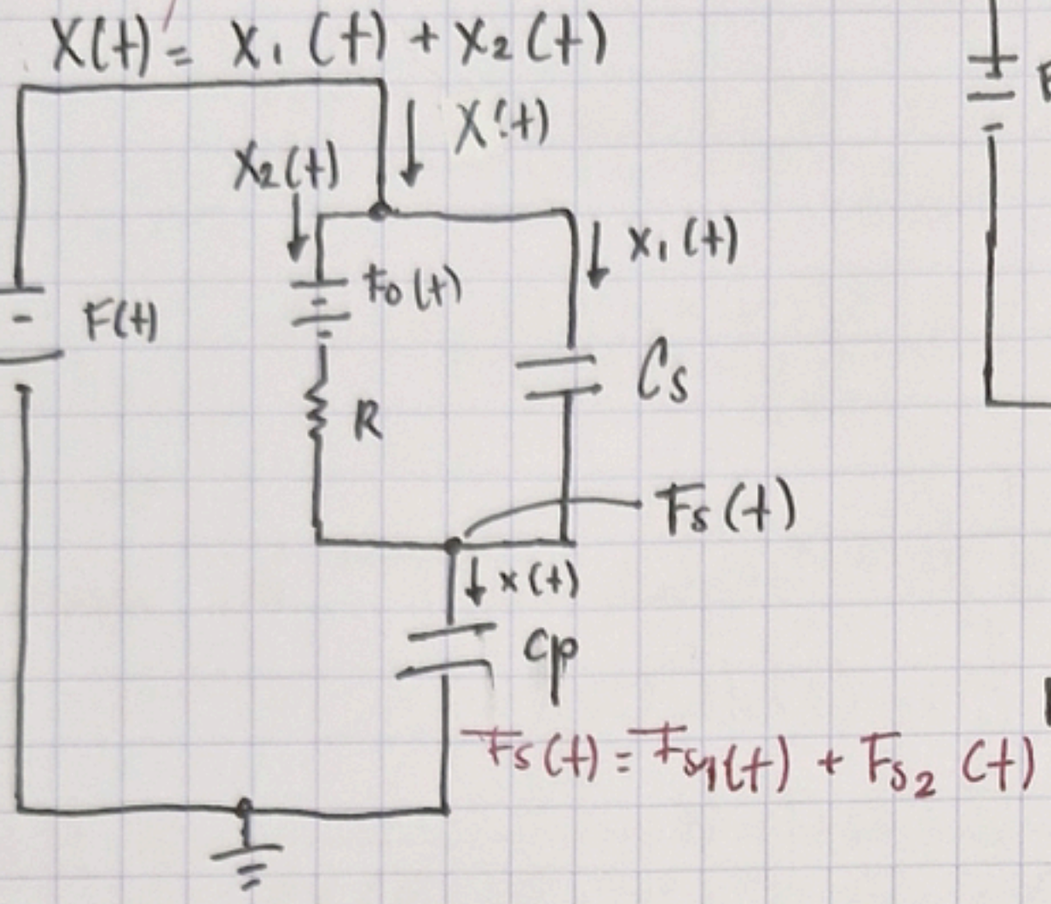
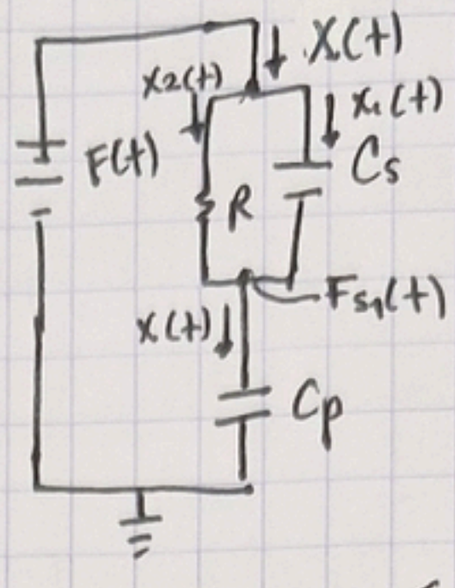
Práctica 3. Modelado y control del sistema

musculoesquelético.

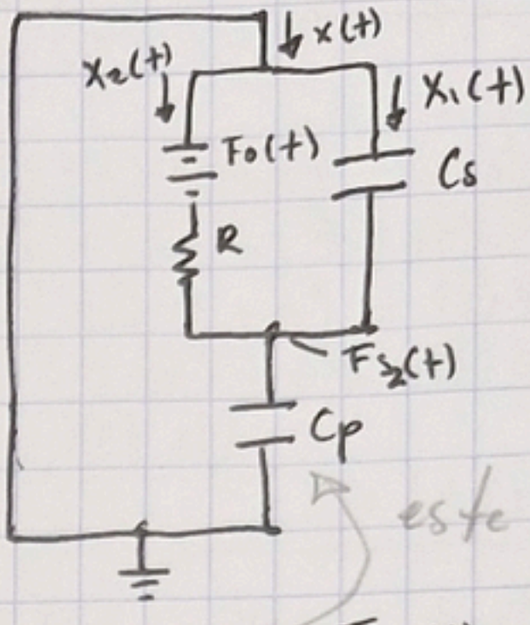


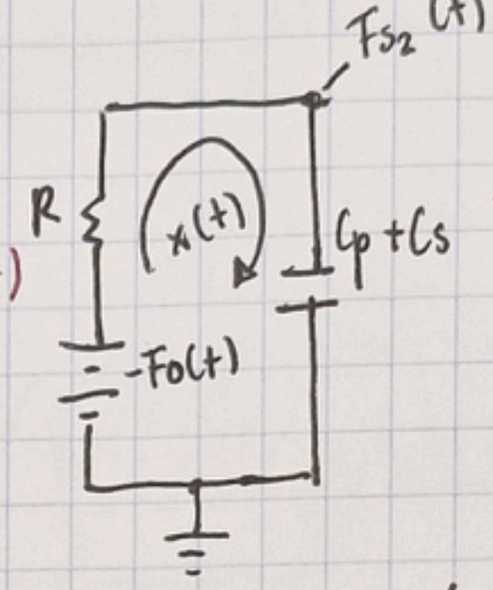
$$F_s(t) = F_{s1}(t) + F_{s2}(t)$$

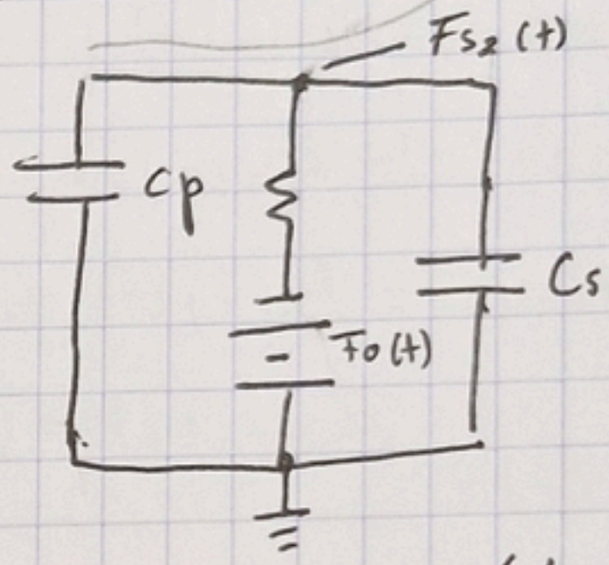
1er cto.



2do cto.







Análisis matemático

Debido a que el sistema contiene dos fuentes de voltaje, se debe aplicar el principio de superposición.

Circuito 1. Apagando $F_o(t)$

$$x(t) = x_1(t) + x_2(t)$$

$$x(t) = C_p \frac{dF_{s1}(t)}{dt}$$

$$x_1(t) = C_s \frac{d[F(t) - F_{s1}(t)]}{dt}$$

$$x_2(t) = \frac{F(t) - F_{s1}(t)}{R}$$

$$C_p \frac{dF_{s1}(t)}{dt} = C_s \frac{d[F(t) - F_{s1}(t)]}{dt} + \frac{F(t) - F_{s1}(t)}{R}$$

$$C_p s F_{s1}(s) = C_s s[F(s) - F_{s1}(s)] + \frac{F(s) - F_{s1}(s)}{R}$$

$$C_p s F_{s1}(s) = C_s s F(s) - C_s s F_{s1}(s) + \frac{F(s)}{R} - \frac{F_{s1}(s)}{R}$$

$$C_p s F_{s1}(s) = C_s s F_{s1}(s) + \frac{F_{s1}(s)}{R} = C_s s F(s) + \frac{F(s)}{R}$$

$$\left(C_p s + C_s s + \frac{1}{R}\right) F_{s1}(s) = \left(C_s s + \frac{1}{R}\right) F(s)$$

$$\frac{R\left((C_p+C_s)s+1\right)}{R}F_{s1}(s)=\frac{C_sR_s+1}{R}F(s)$$

$$F_{s1}(s)=\frac{\frac{C_sR_s+1}{R}}{\frac{R\left((C_p+C_s)s+1\right)}{R}}F(s)$$

$$F_{s1}(s)=\frac{C_sRs+1}{R\left((C_p+C_s)s+1\right)}F(s)$$

Circuito 2: Apagando F(t)

Recordando que $F_o(t)=aF(t)$

$$-aF(t)=Rx(t)+\frac{1}{C_p+C_s}\int x(t)dt$$

$$F_{s2}(t)=\frac{1}{C_p+C_s}\int x(t)dt$$

$$F(s)=-aF(s)=Rx(s)+\frac{x(s)}{(C_p+C_s)s}=-\frac{1}{\alpha}\left(R+\frac{1}{(C_p+C_s)s}\right)x(s)=-\frac{1}{\alpha}\frac{R(C_p+C_s}{\alpha\,(C_p+C_s}\cdot x(s)$$

$$F_{s2}(s)=\frac{x(s)}{(C_p+C_s)s}$$

$$\frac{F_{s2}(s)}{F(t)}=\frac{\frac{x(s)}{(C_p+C_s)s}}{-\frac{1}{\alpha}\frac{R(C_p+C_s)s+1}{(C_p+C_s)s}x(s)}=-\frac{\alpha}{R(C_p+C_s)s+1}$$

$$F_{s2}(s)=\frac{\alpha F(t)}{R(C_p+C_s)s+1}$$

Función de Transferencia

$$F_s(s)=F_{s1}(s)+F_{s2}(s)=\frac{C_sR_s+1}{R(C_p+C_s)s+1}F(s)-\frac{\alpha F(s)}{R(C_p+C_s)s+1}$$

$$F_s(s)=\frac{C_sR_s+1-\alpha}{R(C_p+C_s)s+1}F(s)$$

$$\frac{F_s(s)}{F(s)}=\frac{C_sR_s+1-\alpha}{R(C_p+C_s)s+1}$$

Error en estado estacionario

$$e(s) = \lim_{s\to 0} s F(s)\left(1 - \frac{F_s(s)}{F(s)}\right) = \lim_{s\to 0} s\,\frac{1}{s}\left(1 - \frac{RC_s s + 1 - \alpha}{(RC_p + RC_s)s + 1}\right) = \alpha$$

Estabilidad en lazo abierto

$$(RC_p + RC_s)s + 1 = 0$$

$$\lambda = -\frac{1}{RC_p + RC_s}$$

El sistema es estable.